Mid-Paper Pattern

CLO-1	CLO-2
Theory	Test-cases
week 1 S1	week 5 S2
week 1 S2	week 6 S2
week 2 S1 & S2	week 7 S1
week 3 S1 & S2	week 7 S2
week 5 S1	week 8 S1
	week 8 S2

Test Covergeing

Test coverage

- Statement coverage
- Decision coverage

Statements

Decision Statement
- if-else
- switch
- loops
- branching

↓ change program
*executed order

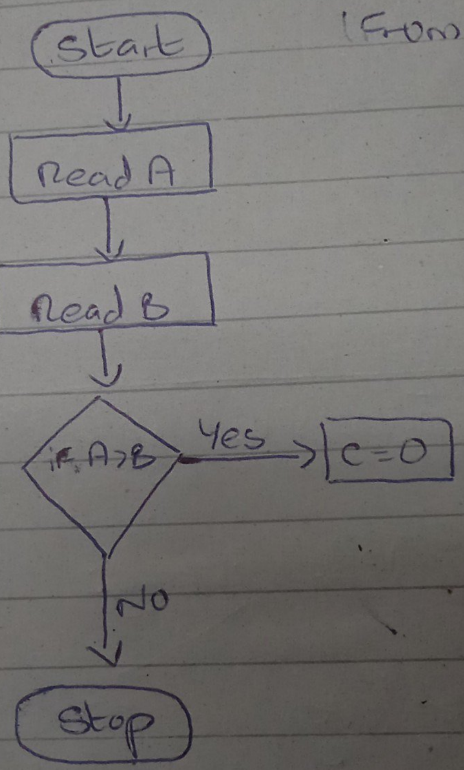
Non-Decision statement
- assignement
- Declaration
- I/O

executed in sequence
doesn't change Program execute order

$$\text{statement coverage} = \frac{\text{No. of } \cancel{\text{student}} \text{ statement exercised}}{\text{Total no. of statements}} \times 100$$

First Step Draw Flow chart :-

(From Image)



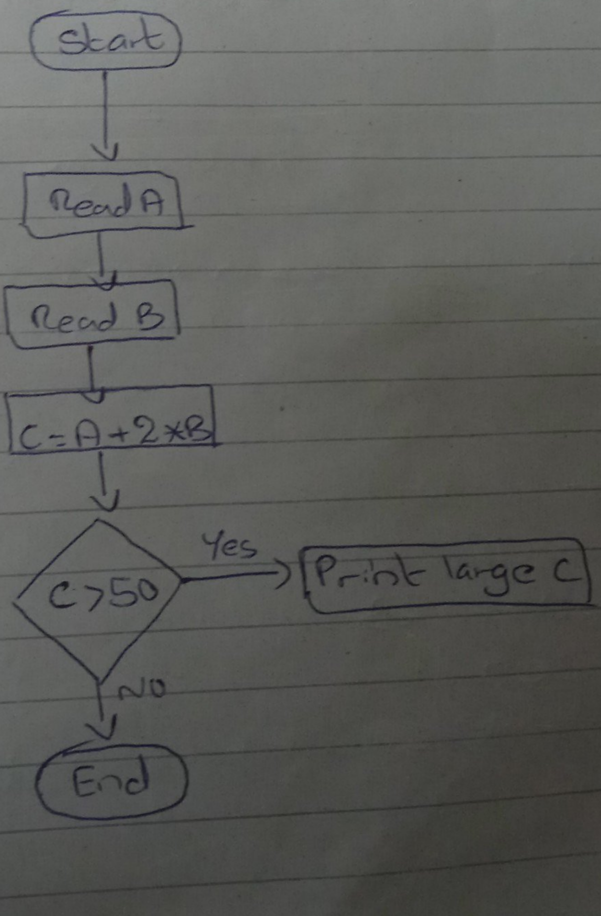
consider c

1. Read A
2. Read B
3. C = A + 2
4. if C > 5
5. Pri
6. End if

exercised
x 200
ments
Image)

consider code.

1. Read A
2. Read B
3. C = A + 2*B
4. if C > 50 then
5. Print large C
6. End if



A	B	C	Line No. of statement covered	No. of statement exercised	Statement covera[ge]
10	20	50	1,2,3,4,6	5	5/6 × 100
20	10	40	1,2,3,4,6	5	5/6 × 100
20	20	60	1,2,3,4,5,6	6	6/6 × 100

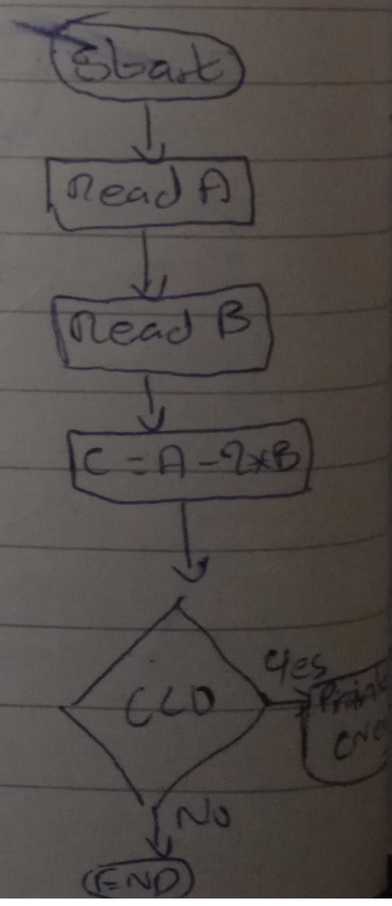
1 test case is required to achieve maximum statement coverage

Decision Coverage

$$\text{Decision coverage} = \frac{\text{No. of decision outcomes exercised}}{\text{Total No. of decision outcomes}} \times 100\%$$

consider code,

```
  ~~Red~~ Read A
  Read B
  C = A - 2*B
4 if C<0 then
5     Print "C Negative"
6 End if
```



A	B	C	Line No. of Path covered
1	2		
10	1		

Statement coverage slide-3 Question of decision coverage

A = Gold card

B = business calass Full

C = Economy Full

A	B	C	Line No. of Path ~~Decision coverage~~	Decision coverage
Yes	Yes	X	1,2,3,5	
NO	Yes	Yes		

Test-1: Gold card holder who gets upgraded to business class.

Test-2: Non-Gold card who stays in economy

Test-3: A person who is bumped from flight

Test-1: 1,2,4,5

Test-2:- 1,6,7,5

Test-3:- 1,6,8,10

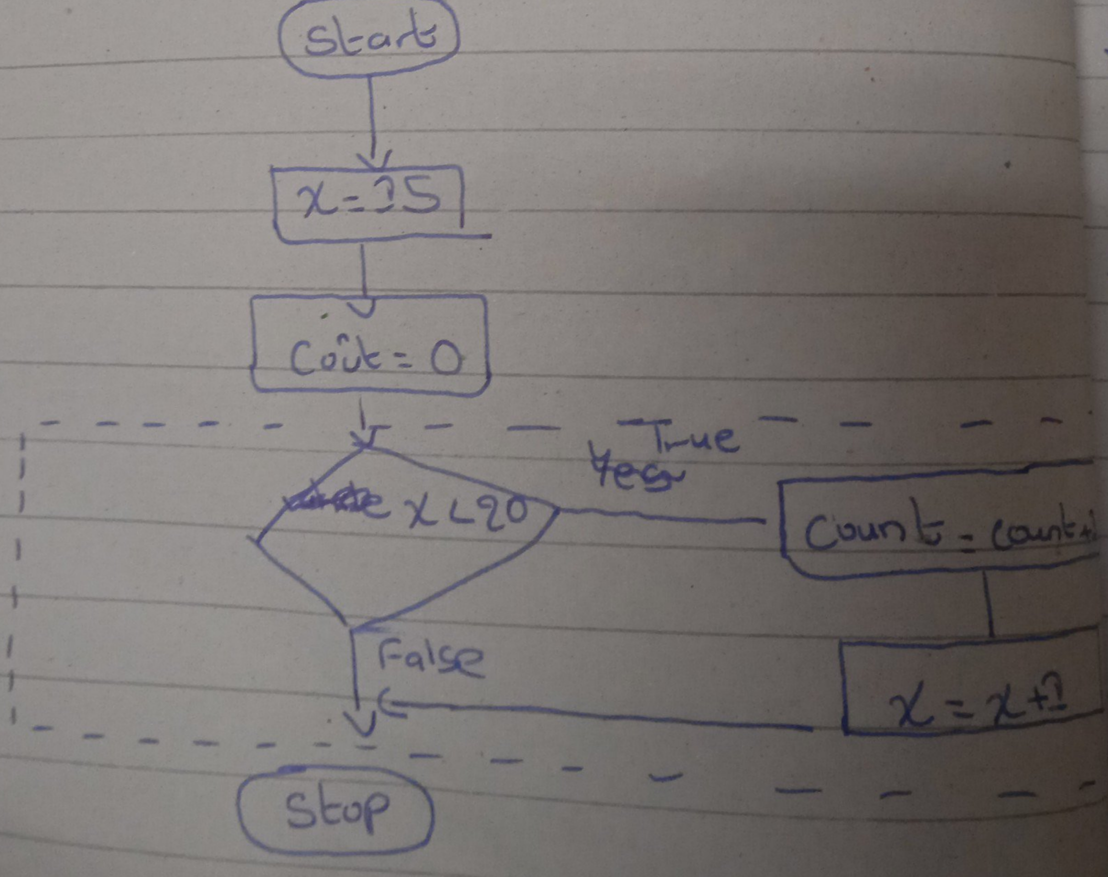
Table From Slide-7

Test	Statement Exercised	Total

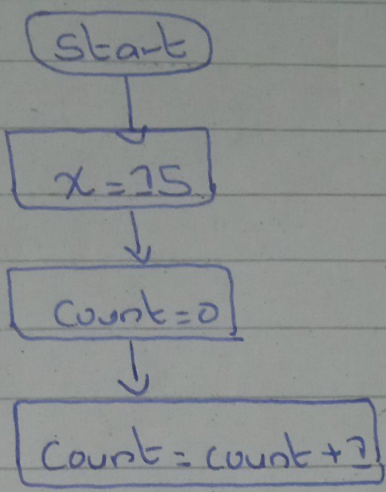
Loops :-

- Loops
  - Pre-test
    - For
    - while
  - Post-test
    - Do while

Loop-1 Flow-chart:- (while)



Loop-2: (Do-while) (Remaining From Image)

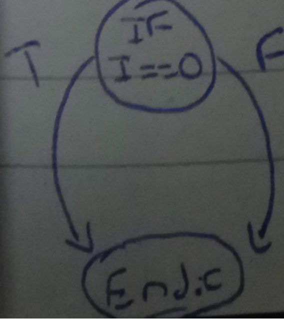


Control Flow Diagram:- Consider decision state

if (i=0)
{
}
else
{
}

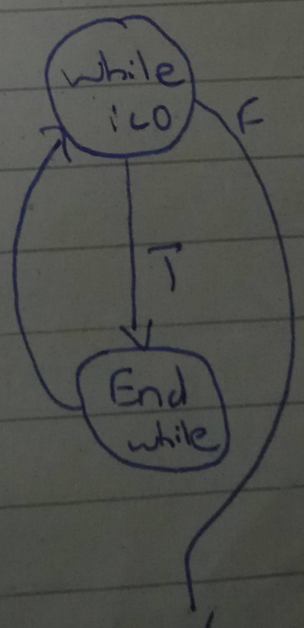
if I==0 THEN
ELSE

CFG



while (i<0) THEN

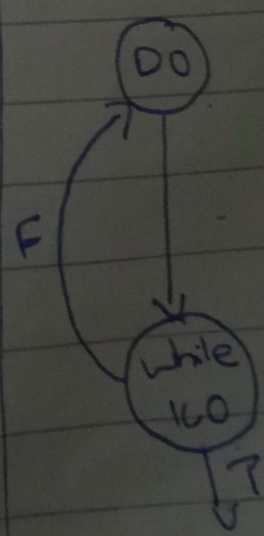
END WHILE



do-while

DO

while (I<0)



Slide-6 code..

First Remove non-executable code..

//

13 IF A>B

14 THEN

IF A>C

THEN

ELSE

END IF

ELSE

IF A≥B>C

THEN

ELSE

END IF

Step-1. Consider outer most if

1

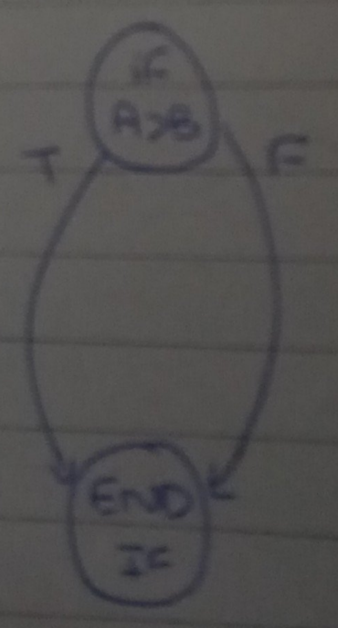
2

3

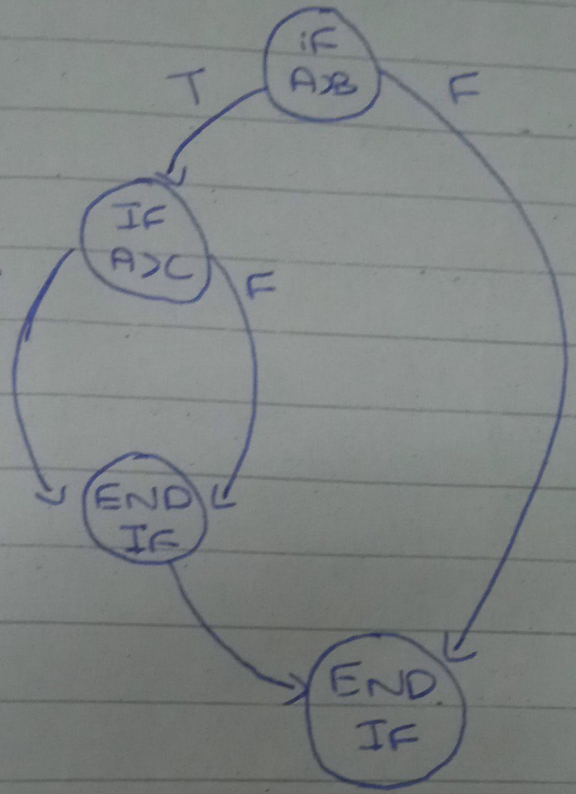
4

5

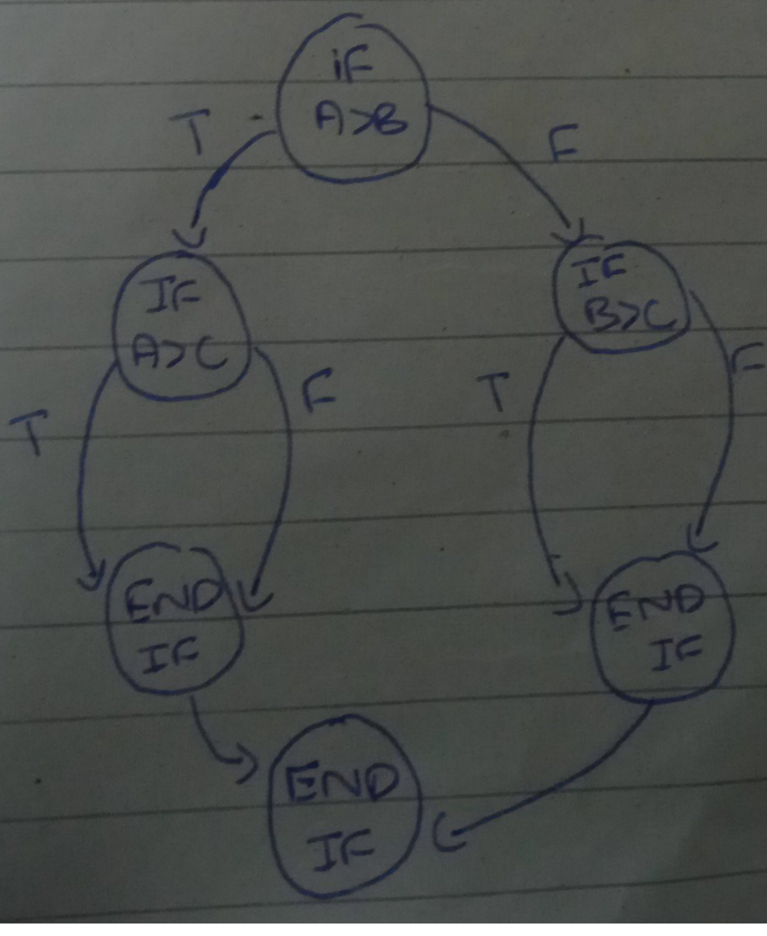
6



Step-2: consider true Path

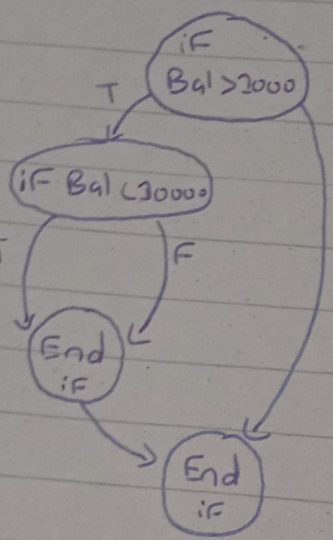


Step-3: consider False Path



HFG Topic:-

Step-1: create CFG of code;



Step-2: Eliminate non executable statements only & rewrite the code & Group non-decision state[ments] in order of their execution sequence.

5. Base Rate = 0.035
6. Interest = Base Rate } 1
8. Read (Balance)
9. IF Balance > 1000
10. THEN
11. Interest = Interest + 0.005 → 2
12. IF Bal < 10000
13. THEN
14. Interest = Interest + 0.05 → 3
15. ELSE

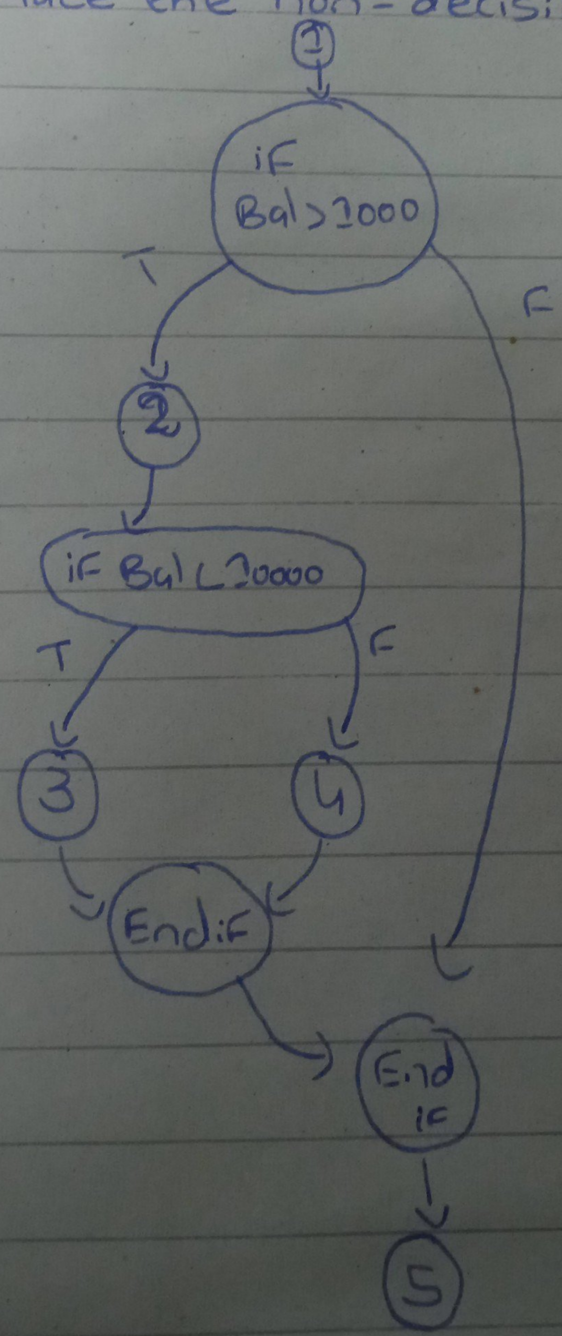
16. Interest = Interest + 0.010 → 4

17. End IF

18. End IF

20. Balance = Balance × (1 + Interest) → 5

Step-3: Place the non-decision groups Accordingly



Q: How many test cases are required to achieve max decisin coverage

Ans: 3 test-cases as total no. of Paths are three

Q: How many test-cases are requred to achieve maximum Statement coverage.

Ans: 2 test-cases are required

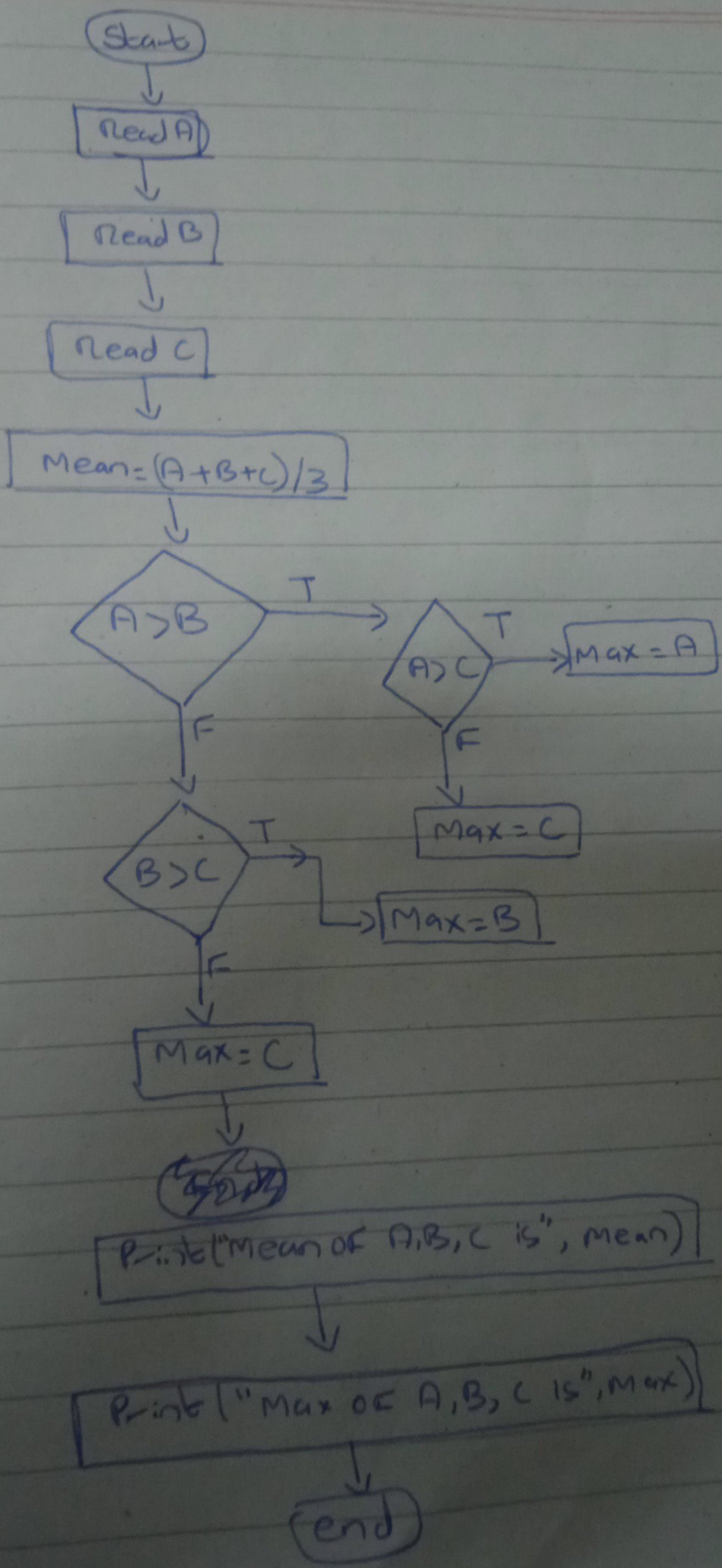
Q: Generate test data for Max statement coverage

Balance:	1001
	10001

Q: Generate test-data for Max decision coverage

Balance	1001
	10001
	999

Slide-5 week-11 S1 one question solve ~~from~~ is in Image.


Start
Read A
Read B
Read C
Mean=(A+B+C)/3
A>B
T
A>C
T
Max=A
F
Max=C
F
B>C
T
Max=B
F
Max=C
Print("Mean of A,B,C is", Mean)
Print("Max of A,B,C is",Max)
end

CLO-3 : week-4

Software Process:-
- steps /Procedure to develop a product
- The activities find expressions in the form of methods, techniques, strategies
- These activities rely on documents, standards and policies.
- The process can be repeated in Subsequent Projects.
- The Process can be evaluated by cost, quality and time to deliver Product.
- Process Improvement is continous cycle.

Process Improvement:
- To improve a process, the organization needs to evaluate its capabilities & limitation
-

Leave Slide 5, 6

CMM - Capability Maturity Level:-

- It is a Framework For evaluating the capabilities of a Software company
- Maturity level tells us what ~~og~~ org. is capable with respect to cost & Quality.

Immature Organization (Slide-8)

Mature Org. (Slide-9)

(Slide-10) Read

Level-1:-

- Software is developed by Following no process
- There is no much planning
- There is no KPA associated

Level-2:-

- At this level, the process exists and is followed
- Performance of Past Project is consider for future Plan.

Scenario question can come like the software house is at level-1 and have to achieve level-2. what will you recommend to achieve the level.

Level-3:-

· Documentation plays key role related to Projec
management & Software development are docume
reviewed & integrated with Organizational Process
· Processes are approved
· Functionality & Quality is tracked.
·

(i) Peer Review:-

Review to find bugs at early by the Same level like ~~or~~ two developers complete the task then check & review each other work.

) Intergroup Cordination:-

cordination between the Software development group with other groups like marketing or SQA.

(iii) Integrated Software Management:-

Integrating the Software with the organizational (business) Activities.

(iv) Training Program:-

Training of the employees.

(i) (ii) (iii) Read from Slide-26

Level-4:-

. At level-4, the quality of Product and Process is measure.

. The metrics then are used to gain quantitative insight.

.2 KPA's (Read slide-27)

Level-5:-

At level-5, the processes are optimized, by the Organization. One option is to ~~fea~~ analyze the level-4 result and optimize the process and the other option is ~~to~~ the change in the methodology or Technology. As a result, the process will be updated.

- Defects Prevention
- Technology change Management
- Process change Management

ISO - Software Quality Standard

It is a global Initiative for understanding the concept of Software quality has been performed.

ISO

- 9126
  - ↓ Product Quality standard
  - ↓ quality characte.
  - ↓ Focus on things visible to user
- 9000:2000
  - ↓ SQA standard
    - 9000
      - ↓ Fundamentals & vocab
      - ↓ 8 Principles
    - 9001
      - ↓ Requirement
      - ↓ 5 kind 7 req.
    - 9004
      - ↓ Perfor Impro...

ISO - 9000:-

- customer Focus to satisfy customer
- Leadership
- Involvement of People
. Org. rely on people (Employees)
-

- Process approach
- System approach to management
- continual Improvement
- Factual approach to decision-making;
  - Decisions may be made based on Facts, experie
  - Facts ~~of~~ can be gathered by using measurement process.

- mutually Beneficial Supplier relationship.

ISO 9001:-

- Remedial Requirements (Data collected ~~by~~ in level-4 of CMM)
- Realization Requirements:- The requirements that transform customer req. into product.
- Resource Requirements:-

-

Systemic Req

↓

QMS (Quality Management System)

↓

ISO 9001: 2000

- Management Requirements:

Week-13:-

Configuration Management:-
The process involved in the development of Software is called configuration Manggement

The items are Source code, test-scrpts, third-party Software, hardware, data

configuration Management has number of important implication for testing

- It allows the testers to manage their test result using the Same configurdtion mechanism.
- configuration management Support the build process which is important for delivery of test realase into test environment.
- It also allows us to map what is being tested to underlying & components that make it up.

- when ve report defects, we need to report Something which is version controlled.

Week-12:-

Test Organization & ~~Tes~~ Independence

Independent testing is carried out by someone other than developer of code being tested. It is hard for authors to identify their own errors but it is easier for others to see them

Benefits:- / Drawbacks (chatgpt)
~~week 11, 12~~ slide 4, 5 (Job Description)

Test Approaches / Strategies:-
- A test approach is the implementation of test strategy on project.
- It defines how testing will be implemented.
- It includes all decisions made on testing based on projects goals as well as risk assessment

Analytical Approach:- Priority / Risk testing.
Model-based:- Testing ~~th~~ on live server by analyzing the Data (Statistical Information)

Standard-compliance

Factors:-

. Before creating a testing approach, analyze your team and create A/c to their skills.

.

Test Planning:-

Test Plan is the important activity undertaken by test leader where the plan is define for the Project. The document created is called Master test Plan or Project test Plan.

Slide-12, 13, Draw template for Test-Plan Section

Week-12 Session 2:-

Test-Estimation:

There are 2 methods of Test Estimation;

(i) Metric based (Based on Past Similar Project da

(ii) Expert based (Based on experience)

Metric-based:-

It relies upon data collected from Previous or Similar Project.

week-12 S2.

- with this approach it is possible to estimate accurately the cost & time req. For similar Project
- It is imp. that actual costs and time for testing are accurately recorded.

Expert-based:-

It is also known as wide Band Delphi approach.

Remaining From GPT

Slide-10: Design a template For Test Reporting

week-14

Testing Maturity Model (TMM)

TMM gives guidance concerning how to improve test process.

There are 5 levels where each level have Maturity goals except level-1. which are achieved by means of ATR's.

Level-1:-

- no maturity goals are specified
- It is called Initial level
- The condition to be at level-1 is that the testing of Project happens done by developer.

week-14 slide 10 next class

Level-2:-

At level-2, Maturity goals are as Follows;

. Develop testing and debugging goals.

. Initiate a testing plan

. Institutionalize basic testing techniques & methods